

Summary of Backpropagation-Free Training

Geng Tian

May 5, 2025

This week, I first designed and trained a VGG16 model using the BP-free method. Subsequently, I attempted to improve the model accuracy by modifying the loss calculation rule. However, these experiments did not proceed smoothly. Upon re-examining the code logic, I realized that if no noise is added to the client-side model, the accuracy of the model should theoretically be lower than that achieved with the original BP-free approach.

1 BP-Free Training of VGG16

I implemented the VGG16 by referring to the open-source code provided in [1]. The decision to use VGG16 instead of ResNet18 was based on previous experimental results: VGG8 consistently outperformed ResNet5, and VGG16 has a simpler structure compared to ResNet18. However, according to the settings used in earlier implementations, training VGG16 using the BP-free method requires an excessively large amount of memory, exceeding the total capacity available on the GPUs I have access to. To address this issue, I began exploring to trade computation time for reduced memory usage. As a result, the current training process is still ongoing and I have not yet obtained the training results.

2 Updating the Loss Calculation Method

I modified the loss computation by applying Equation 1, where each data sample is processed N times in a single epoch. For each $i \in N$, loss_i represents the standard loss during the i -th forward pass.

$$\hat{\text{loss}}_i = \text{loss}_i - \sum_{i=1}^N \frac{\text{loss}_i}{N} \quad (1)$$

Despite this change, no improvement in model accuracy was observed. The comparison between the old and new loss calculation methods is summarized in Table 1.

Table 1: Model accuracy under different loss computation strategies.

Loss Computation Strategy	Model	Accuracy (%)
Standard Loss	ResNet-5	58.48
	VGG-8	73.17
Multi-pass Mean-Centered Loss	ResNet-5	58.26
	VGG-8	72.69

3 Model Accuracy: Intuitive Analysis

Our method aims to reduce memory consumption, computational load, and communication overhead simultaneously. To achieve this, the client model performs only one forward pass per mini-batch, while the server-side model may undergo thousands of forward passes within the same batch. This imbalance results in minimal updates to the client model, which ultimately limits the overall model accuracy. Therefore, it is reasonable to expect that the accuracy of our method will be inherently lower than that of the standard BP-free training approach.

References

- [1] J. Jiang, Z. Zhang, C. Xu, Z. Yu, and Y. Peng, “One forward is enough for neural network training via likelihood ratio method,” *arXiv preprint arXiv:2305.08960*, 2023.